Dr. Sergio Ruocco

Embedded and real-time systems, Linux and toolchain consultant · Senior Software Engineer at EMASS
EMASS (an NTU/A*STAR spin-off), Singapore

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Profile

Dr. Sergio Ruocco is an embedded-systems and toolchain consultant with 30 years of experience in Linux kernel and driver development, real-time operating systems, compiler toolchains, and non-volatile memory. He helps teams de-risk unfamiliar hardware and software: porting operating systems and SDKs to new SoCs, writing and debugging kernel drivers, building GCC backends, improving real-time behaviour, analysing system performance, and setting up CI/CD build and test pipelines and IT/DevOps infrastructure. He is currently a Senior Software Engineer at EMASS, an NTU/A*STAR spin-off, where he builds the full software stack, leads the Zephyr RTOS port with a team he recruited and mentors, and runs the CI/CD and IT/DevOps infrastructure for the ECS-DoT ultra-low-power RISC-V AI SoC. His clients have included Seagate (USA) and Micron Italy, and he has led EU ENIAC projects with STMicroelectronics.

Capabilities

Engagements range from architecture reviews and feasibility studies to hands-on porting, driver development and debugging.

Operating systems and platforms. Linux (Debian, Ubuntu, Mint, Red Hat) on workstations and datacentre servers; L4 microkernels, MbedOS, Zephyr, RTOS internals and porting; Linux kernel, memory management and device drivers; ZFS and zfs-root workstations and storage servers; IPMI, RAID, GPU servers; Secure build systems and export-controlled software licence servers.

Languages and toolchains. C, GCC/G++, GDB and the GNU toolchain, native and cross-compiled; GCC backend and cross-compiler development; Buildroot, Android SDK, MbedOS SDK; Python, Bash, Git, SSH, GPG, tmux/screen; AI-assisted coding; Docker and Buildx, GitLab CI multi-project pipelines, GitHub and GitLab container registries; IT/DevOps infrastructure for engineering teams.

Architectures and hardware. ARM and RISC-V platforms; board bring-up, JTAG and RISC-V debug (OpenOCD); Microcontroller firmware on RP2040/RP2350: Pico SDK, PIO, TinyUSB, I²C and UART; FPGA boards, peripherals, memory maps and I/O devices; Non-volatile memory (NVM-DIMM, PCM, STT-MRAM) and storage systems; AI accelerators and video-compression hardware.

Methods. Real-time scheduling and temporal reflection; Performance analysis and system tracing; Debugging poorly documented hardware and software; Technical leadership, hiring and mentoring, procurement and project delivery.

Selected projects

Full software stack for the ECS-DoT ultra-low-power RISC-V AI SoC

*2024 – present · NTU, then EMASS (an NTU/A*STAR spin-off) from 06/2025*

Research and develop the device drivers, RTOS, SDK and toolchains needed to compile, optimise and run AI applications on ECS-DoT, a custom ultra-low-power SoC with a RISC-V core and dedicated hardware accelerators, designed and fabricated by the team.

Impact: Enables AI workloads on the ECS-DoT SoC; builds on the AI-accelerator driver and RTOS work done at A*STAR I²R.

ECS-DoT boot agent and JTAG adapter on the Raspberry Pi Pico 2

2026 – present · EMASS · ECS-DoT development boards

Extended the open-source DirtyJTAG firmware into the boards' boot agent: it loads RISC-V applications into the SoC over JTAG at power-up, serves as an OpenOCD adapter and USB-serial bridge, and drives the boards' power control and measurement.

Impact: Shipped to customers in tagged firmware releases since 07/2026.

MbedOS and SDK port to the ECS-DoT RISC-V SoC

*2020 – 2024 · A*STAR I²R / NTU · ECS-DoT Edge SoC with AI acceleration*

Ported the ARM MbedOS 5.12.4 RTOS and its SDK to the ECS-DoT SoC, covering a different CPU core, instruction set, peripherals, memory map and I/O devices. Both the original and the target platforms were poorly documented and buggy, so the work was a complex trial-and-error effort with the GCC toolchain and GDB.

Impact: Delivered a working RTOS port and board-support package; technology disclosure SWIP-PMEP-2022-31, licensed to EMASS Pte Ltd.

AI accelerator Linux kernel driver port to MbedOS on RISC-V

*2022 – 2023 · A*STAR I²R · ECS-DoT*

Ported the AI accelerator's ARM Linux kernel driver to MbedOS on the ECS-DoT RISC-V CPU, adapting the driver to a new operating system, memory map and device model.

Impact: Technology disclosure SWIP-PNVD-2022-33, licensed to EMASS Pte Ltd.

Real-time extensions and reflective scheduling for the L4 microkernel

2004 – 2007 · NICTA · ERTOS Programme

Analysed the real-time behaviour of general-purpose L4 microkernels, identified the architectural obstacles to predictable scheduling, and designed and implemented a reflective real-time scheduler that lets a system observe and control its own temporal behaviour.

Impact: Published at IEEE RTSS 2006 and OSPERT 2006/2007 and in the EURASIP Journal on Embedded Systems; the work influenced the real-time features of the commercial OKL4 microkernel.

Linux memory-management extensions for non-volatile DIMMs

*2015 · A*STAR DSI · CERN OpenLab collaboration*

Extended Linux memory management to support non-volatile DIMMs, allowing persistent data to be placed directly in main memory and processed at memory speed.

Impact: Procured an invitation to collaborate with CERN OpenLab and an invited talk at CERN OpenLab 2015.

GCC backend for a custom 16-bit microcontroller

2009 – 2010 · Micron Italy (formerly STMicroelectronics) · Università di Milano-Bicocca

Ported the GNU GCC C compiler to a custom, non-standard 16-bit microcontroller managing embedded NAND/NOR phase-change memories under hard real-time constraints and security requirements. The target was poorly documented, requiring a new backend and extensive debugging with GDB.

Impact: Delivered under a Micron R&D contract; the compiler supported devices deployed in high volume.

Active Discs Clusters Performance Study (ADCPS)

*2017 – 2019 · A*STAR DSI, then I²R · funded by Seagate USA*

Industry-funded performance study of active disc clusters, covering workload characterisation, measurement and analysis.

Impact: Delivered a performance study to Seagate USA.

EU ENIAC SMART and TOISE projects

2009 – 2013 · Università di Milano-Bicocca · Micron Italy and STMicroelectronics

Principal Investigator and technical lead for UNIMIB in two three-year EU ENIAC projects: SMART (techniques and tools for advanced real-time embedded firmware, €302,400) and TOISE (trusted computing for European embedded systems, €556,000). Responsible for planning, budget, recruiting research engineers and post-docs, and scientific and technical delivery.

Impact: Funded a research fellowship and a team of post-docs and research engineers; produced tools and techniques for advanced embedded firmware.

Experience

Senior Software Engineer — EMASS (an NTU/A*STAR spin-off)

06/2025 – present · Singapore

End-to-end software for ECS-DoT, an ultra-low-power AI SoC with a RISC-V core and dedicated hardware accelerators. Research and development of the full stack of software and tools — device drivers, RTOS, SDK and toolchains — to compile, optimise and run AI applications on a custom SoC designed and fabricated by the team.

Research Fellow — School of Computer Science and Engineering, Nanyang Technological University (NTU)

04/2024 – 05/2025 · Singapore

Kept the ECS-DoT system-software engineering going between A*STAR I²R and EMASS, laying the groundwork for the EMASS role: the first bare-metal SDK, its SPI and camera drivers, and analysis of hardware bugs in the SoC's SPI peripheral.

Senior Scientist II — A*STAR Institute for Infocomm Research (I²R)

06/2018 – 09/2023 · Singapore

Embedded real-time operating systems, SDKs, firmware and drivers for a RISC-V-based SoC customised with a deep-learning accelerator for ultra-low-power edge-IoT applications.

Scientist II — A*STAR Data Storage Institute (DSI)

10/2012 – 05/2018 · Singapore

Systems software and programming-toolchain research for next-generation solid-state storage and non-volatile memory.

Research Fellow — Dipartimento di Informatica, Sistemistica e Comunicazione (DISCo), Università di Milano-Bicocca
04/2007 – 09/2012 · Milan, Italy
Research and teaching in real-time embedded systems, compiler retargeting and toolchains, and next-generation non-volatile storage, with projects funded by industry contracts and European research grants.

Researcher, Embedded, Real-Time and Operating Systems (ERTOS) Programme — NICTA (National ICT Australia, now Data61/CSIRO), Sydney
04/2004 – 03/2007 · Australia
Real-time and operating-systems research on the L4 microkernel.

Linux Kernel Engineer — Comerson s.r.l., Milan
2002 – 2004 · Italy
Smart video camera with TCP/IP support — at a time when other digital webcams were serial or USB.

Software Engineer — Metamatica, Milan
2000 – 2001 · Italy
Web-based lottery game.

Embedded Software Engineer — CD Express
1996 – 1997 · Italy
MPEG decoder for the Commodore CD-32.

Software Engineer and Independent Consultant — Self-employed
01/1990 – 01/2002 · Italy
Consulting on embedded real-time systems while attending university.

Patents and technology disclosures

*EMASS (Embedded A.I. Systems Pte Ltd) is a Singaporean fabless semiconductor company founded in 2020 as a spin-off from NTU Singapore and A*STAR research, and acquired by Nanoveu Ltd (ASX: NVU) in March 2025.*

SWIP-PMEP-2022-31 — Porting of MbedOS 5.12.4 to the ECS-DoT SoC and RISC-V CPU. Filed 2023-03-10. Review completed 27 April 2023. Licensed to EMASS Pte Ltd.

SWIP-PNVD-2022-33 — Porting of the AI accelerator ARM Linux kernel driver for MbedOS on the ECS-DoT RISC-V CPU. Filed 2023-03-10. Review completed 2 May 2023. Licensed to EMASS Pte Ltd.

SWIP-TLCN-2022-32 — Tracing Linux compiler and kernel driver for an AI accelerator. Filed 2023-03-10. Review completed 21 April 2023. Licensed to EMASS Pte Ltd.

TD2022128 / I2R/Z/13931 — Edge SoC with AI acceleration. Filed 2023-01-12. Licensed to EMASS Pte Ltd.

Publications

Selected publications.

1. Q. Trinh, S. Ruocco, M. Alioto, “Time-Based Sensing for Reference-Less and Robust Read in STT-MRAM Memories,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(10):3338–3348. DOI
2. Q. Trinh, S. Ruocco, M. Alioto, “Novel Time-Based Sensing Scheme for STT-MRAMs,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, Florence, Italy, 1–5. DOI
3. S. Ruocco, D.-K. Le, “Efficient Persistence of Financial Transactions in NVM-based Cloud Data Centers,” *International Conference on Cloud Computing Research and Innovation (ICCCRI)*, 25–36. DOI
4. S. Ruocco, “A Real-Time Programmer’s Tour of General-Purpose L4 Microkernels,” *EURASIP Journal on Embedded Systems*, 2008:1–14. DOI
5. S. Ruocco, “User-Level Fine-Grained Adaptive Real-Time Scheduling via Temporal Reflection,” *27th IEEE Real-Time Systems Symposium (RTSS)*, Rio de Janeiro, Brazil, 246–256. DOI

Education

2004 — PhD in Computer Science
Università degli Studi di Milano, Italy — Mathematics, Statistics, Computational Sciences and Computer Science Research
Thesis: *Temporal Reflection*. Doctoral scholarship, 2000–2004. In Italy, PhD degrees are pass/no-pass and are not graded.

1999 — Master of Science in Computer Science
Università degli Studi di Milano, Italy — 110/110 cum laude
Thesis: *HyperReal: a paradigm for the design and synthesis of hard real-time control systems*.